

VoIP SDK - SDVoE

Revision History

Date	Version	Revision
2022.01.19	2.16	VOIPS.dll - Add function : SET_HTTP_PORT - Add class : WALL_PARAMETER - Add class : MULTIVIEW_PARAMETER - Add class : BEZEL_PARAMETER - Add class : VIDEO_INFO - Modified class : MULTI_LINK_RECEIVER_NODE_CONFIG - Modified class : MULTI_LINK_RECEIVER_NODE_STATUS - Modified enum : DISPLAY
2022.03.14	2.17	VOIPS.dll - Fixed function : GetResult - Fixed & modified function : SET_MULTI_VIEW Example Project - Modified “Dual Link Mode” page - Modified “Multiview” page - Fixed some combobox list
2022.07.15	2.20	VOIPS.dll - Fixed function : SET_MULTI_VIEW - Add function : SET_MULTIVIEW Example Project - Modified : QueryEvent(Type) - Add : SET_MULTIVIEW Button

Date	Version	Revision
2022.08.05	2.21	VOIPS.dll - Add function : DiscoveryIdentity - Add function : GET_SN - Add function : LIST_MULTICAST - Modified class: RESULT - Add class : Multicast - Add function : ROUTE_IO_RS232_BROADCAST - Add function : ROUTE_IO_INFRARED_BROADCAST - Add function : TX_VIDEO_SOURCE Example Project - Add : Discovery Identity Button - Add : GET_SN Button - Add : LIST_MULTICAST Button - Add : ROUTE_IO_RS232_BROADCAST Button - Add : ROUTE_IO_INFRARED_BROADCAST Button
2022.08.15	2.22	VOIPS.dll - Fixed function : GET_SN
2022.11.14	2.23	VOIPS.dll - Add constructor : VOIPS(string Address, [int Port = 80]) - Add function : SET_CONTROL_SERVER Example Project - Add : SET_CONTROL_SERVER Button - Modified : GET_ALL_SETTINGS
2022.11.17	2.24	VOIPS.dll - Add function : GET_MCU_IP - Modified function : GET_SN Example Project - Add : GET_MCU_IP Button
2022.12.01	2.25	VOIPS.dll - Add function : GET_MCU_VERSION Example Project - Add : GET_MCU_VERSION Button
2023.03.15	2.26	VOIPS.dll - Fixed function : SET_VIDEO_WALL Example Project - Modified "Video Wall" Page

Date	Version	Revision
2023.05.24	2.27	VOIP.dll - Add function : RX_HDMI_AUDIO_SOURCE - Add function : TX_SET_SCALER
2023.08.08	2.27	VOIP.dll - Add function : UART_SEND
2023.09.08	2.28	VOIP.dll - Modified function : SET_MULTIVIEW - Rename function : RX_MULTICH_AUDIO_SOURCE-> MULTICH_AUDIO_SOURCE
2023.12.12	2.29	VOIP.dll - Fixed & Modified function : ADD_DEVICE_ALL Example Project - Modified "ADD_DEVICE_ALL"
2024.06.07	2.30	VOIP.dll - Fixed function : ICRON_USB QUERY - Fixed function : GET_ALL_SETTINGS - Modified Enum : NODE_TYPE

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1. INTRODUCTION

This VoIP (Video over IP) SDK is a powerful and flexible solution for controlling multiple SDVoE (Software Defined Video over Ethernet) based extenders within same network. The user only needs to program this SDK into PC within the same local network as the extenders (transmitters and receivers) to define and configure channel routing selections (including video, audio, and a variety of control interface types). Without the use of this centralized control SDK, each transmitter and receiver pair would only be able to function in a point-to-point capacity.

Additionally, this SDK supports controlling and configuring the matrix, video wall, and multiviewer modes of connected SDVoE units. The settings of all connected transmitter/receiver units, including IP configuration, compatibility settings, and extender status are listed in this document.

Note. The usage of SDK only be programmed on MS OS, the MAC OS is not supported.

1.1 File List

File	Version
VOIPS.dll	V2.30
Example Project	V2.30
Blueriver Control Server	V3.3.0.0 V3.5.0.0 V3.7.0.0

1.2 Blueriver Considerations

(1) When using VOIPS.dll, need to execute “controlserver.exe” before running any program.

(2) Make sure “controlserver.exe” have internet access permission if firewall is enabled.

Note. Please consult to your MIS administrator or local network service if there’s any issue on firewall or network access permission.

2. Class VOIPS

Supported devices: SDVoE related products

2.1 Enum

2.1.1 enum VIDEOWALL_ASPECT_RATIO

No	Name	Description
0	eFull	Specifies the videowall aspect ratio as full screen.
1	eBest_Fit	Specifies the videowall aspect ratio as best fit.
2	e4_3	Specifies the videowall aspect ratio as 4:3.
3	e16_9	Specifies the videowall aspect ratio as 16:9.

2.1.2 enum DEVICE_GROUPS

No	Name	Description
0	ALL	Specifies the unit status as Transmitter and Receiver at the same time.
1	TRANSMITTER	Specifies the unit status as Transmitter.
2	RECEIVER	Specifies the unit status as Receiver.

2.1.3 enum TIMING

No	Name	Description
0	e1024x768p60	Specifies timing as 1024x768p60
1	e1280x768p60	Specifies timing as 1280x768p60
2	e1280x960p60	Specifies timing as 1080x960p60
3	e1280x1024p60	Specifies timing as 1280x1024p60
4	e1360x768p60	Specifies timing as 1360x768p60
5	e1400x1050p60	Specifies timing as 1400x1050p60
6	e1600x1200p60	Specifies timing as 1600x1200p60
7	e1680x1050p60	Specifies timing as 1680x1050p60
8	e1920x1200p60	Specifies timing as 1920x1200p60
9	e720p50	Specifies timing as 720p50
10	e720p60	Specifies timing as 720p60
11	e1080p50	Specifies timing as 1080p50
12	e1080p60	Specifies timing as 1080p60
13	e3840x2160p25	Specifies timing as 3840x2160p25

14	e3840x2160p30	Specifies timing as 3840x2160p30
15	e3840x2160p50	Specifies timing as 3840x2160p50
16	e3840x2160p60	Specifies timing as 3840x2160p60
17	e4096x2160p24	Specifies timing as 4096x2160p24
18	e4096x2160p30	Specifies timing as 4096x2160p30
19	e4096x2160p60	Specifies timing as 4096x2160p60
20	eEnd	None

2.1.4 enum DISPLAY

No	Name	Description
0	NONE	
1	GENLOCKED	Specifies video mode is to be set to genlock (i.e. synchronized zero-frame latency) full screen 1:1 display mode.
2	GENLOCK_SCALING	Specifies the video mode is to be set to genlock (i.e. zero-frame latency) full screen mode with scaling.
3	FAST_SWITCHED	Specifies the video mode is to be set to fast switched full screen mode, with aspect ratio preserved by adding black bars when needed. Note: When using the API with an AVX device, when a JSON return indicates the status of FASTSWTICH, the device is functioning in the FORCED OUTPUT RESOLUTION mode (applicable to AVX200T devices only).
4	FAST_SWITCHED_CROP	This string specifies the video mode is to be set to fast switched full screen mode, with aspect ratio preserved by cropping the video.
5	FAST_SWITCHED_STRETCH	This string specifies the video mode is to be set to fast switched full screen mode, with the video stretched to cover the entire screen without regard for preserving the aspect ratio.
6	WALL_FAST_SWITCHED	Specified the video mode is to be set to fast switched display wall mode.
7	WALL_GENLOCKED	Specifies the video mode is to be set to genlocked (synchronized) display wall mode.
8	MULTIVIEW	multiview mode.

2.1.5 enum EVENT_TYPES

No	Name	Description
0	DEVICE_CONNECTED	The <code>DEVICE_CONNECTED</code> event indicates that a new

		device has joined the network.
1	DEVICE_DISCONNECTED	The DEVICE_DISCONNECTED event indicates that a device has left the network.
2	DEVICE_INFO_AVAILABLE	The DEVICE_INFO_AVAILABLE event indicates that device information is available for a newly discovered device.
3	REQUEST_COMPLETE	REQUEST_COMPLETE event indicates completion of a background command, response status PROCESSING and a request ID instead of status SUCCESS.
4	SETTINGS_CHANGED	SETTINGS_CHANGED event indicates that new settings are returned.
5	I2C	I2C event indicates that an I2C bus master (e.g. a MCU) has generated an event through the I2C interface.
6	CEC_RECEIVED	CEC_RECEIVED event indicates that Consumer Electronics Control (CEC) data received by a specific device.
7	RS232_RECEIVED	RS_232_RECEIVED event indicates RS-232 data received by a specific device.
8	INFRARED_RECEIVED	INFRARED_RECEIVED event indicates Infrared data received by a specific device.
9	FIRMWARE_UPDATE_START	Firmware update started. No operations can be performed on device until update done.
10	FIRMWARE_UPDATE_PROGRESS	Firmware update of device in progress.
11	FIRMWARE_UPDATE_COMPLETE	Firmware update of device has completed.

2.1.6 enum IP_MODE

No	Name	Description
0	STATIC	Internet Protocol configuration is fixed and configured manually.
1	DHCP	Internet Protocol configuration is obtained from a DHCP server, with fallback on auto-IP if no DHCP server is available.
2	UNKNOWN	

2.1.7 enum BAUDRATE

No	Name	Description
0	BAUDRATE_2400	2400 bit/s
1	BAUDRATE_4800	4800 bit/s
2	BAUDRATE_9600	9600 bit/s

3	BAUDRATE_19200	19200 bit/s
4	BAUDRATE_38400	38400 bit/s
5	BAUDRATE_57600	57600 bit/s
6	BAUDRATE_115200	115200 bit/s
7	BAUDRATE_UNKNOWN	Specific the unknown baudrate detected

2.1.8 enum DATABIT

No	Name	Description
0	DATABIT_7	7 bits
1	DATABIT_8	8 bits
2	DATABIT_UNKNOWN	Specific the unknown databit detected

2.1.9 enum PARITY

No	Name	Description
0	PARITY_NONE	No parity check occurs.
1	PARITY_ODD	Sets the parity bit so that the count of bits set is an odd number.
2	PARITY_EVEN	Sets the parity bit so that the count of bits set is an even number.
3	PARITY_UNKNOWN	Specific the unknown parity detected

2.1.10 enum STOPBIT

No	Name	Description
0	STOPBIT_1	One stop bit
1	STOPBIT_2	Two stop bits
2	STOPBIT_UNKNOWN	Specific the unknown stopbit detected

2.1.11 enum GPIO_MODE

No	Name	Description
0	INPUT	Set GPIO pin as an input.
1	OUTPUT	Set GPIO pin as an output.

2.1.12 enum USB_ROLE

No	Name	Description
0	DISABLED	
1	OTG	The device automatically detects the type of the connected USB HID device and set its role to LOCAL or REMOTE accordingly.
2	LOCAL	The device acts as USB device that connects to a USB host (e.g. a PC).
3	REMOTE	The device acts as USB host to which USB HID devices connect (keyboard, mouse,

		etc.)
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2.1.13 enum NODE_TYPE

No	Name	Description
0	STEREO_AUDIO_INPUT	Stereo Audio Input Node
1	STEREO_AUDIO_INPUT_OUTPUT	Stereo Audio Input /Output Node
2	STEREO_AUDIO_OUTPUT	Stereo Audio Output Node
3	BUTTON	Button Node
4	CEC	CEC Node
5	COLOR_GENERATOR	Color Generator Node
6	FRAME_BUFFER	Frame Buffer Node
7	FRAME_RATE_CONVERTER	Frame Rate Converter Node
8	GPIO	General-Purpose Input-Output (GPIO) Node
9	HDMI_DECODER	HDMI Decoder Node
10	HDMI_ENCODER	HDMI Encoder Node
11	HDMI_MONITOR	HDMI Monitor Node
12	MULTICH_AUDIO_INPUT	Multi Channel Audio Input Node
13	MULTICH_AUDIO_OUTPUT	Multi Channel Audio Output Node
14	INFRARED_DECODER	Infrared Decoder Node
15	INFRARED_ENCODER	Infrared Encoder Node
16	LED	LED Node
17	NETWORK_INTERFACE	Network Interface Node
18	NETWORK_PORT	Network Port Node
19	NETWORK_SWITCH	Network Switch Node
20	NULL_SOURCE	Null Source Node This node is used as an input source to other nodes to represent the fact that no valid input is available.
21	SCALER	Scaler Node
22	UART	UART Node
23	USB_HID	USB Human Interface Device (HID) Extender Node
24	USB_ICRON	Icron USB Extender Node
25	VIDEO_INPUT	Video Input Node
26	VIDEO_COMPRESSOR	Video Compressor Node
27	VIDEO_DECOMPRESSOR	Video Decompressor Node
28	BITMAP_OVERLAY	Bitmap Overlay Node This node is only applicable to endpoint devices that include an NT2000 chipset only.

29	THUMBNAIL_GENERATOR	Thumbnail Generator Node Applicable only the endpoint devices containing a NT1000 or NT2000 chipset.
30	MULTI_LINK_TRANSMITTER	
31	MULTI_LINK_RECEIVER	
32	USB_EXTENDER	
33	UNKNOWN	Unknown node type

2.1.14 enum REF_TYPE

No	Name	Description
0	HDMI	HDMI Video stream
1	SCALER	Scaler Video stream

2.1.15 enum AUDIO_STREAM

No	Name	Description
0	HDMI_AUDIO	HDMI Audio stream input output
1	STEREO_AUDIO	Stereo Audio Stream input output
2	MULTICH_AUDIO	Multichannel Audio Stream input output

2.1.16 enum MULTI_LINK_MODE

No	Name	Description
0	SINGLE	Single Mode
1	DUAL	Dual Link Mode

2.1.17 enum ICRON_USB_ROLE

No	Name	Description
0	LOCAL	Local USB Device
1	REMOTE	Remote USB Device

2.2 Class

2.2.1 Class BR_RESULT

Type	Name	Description
string	status	Possible results for status are: SUCCESS: Command succeeded. ERROR: Command failed. PROCESSING: Command executed as a background command. Possible to execute other commands while waiting for result.
Int?	request_id	A unique request ID is assigned to each command executed as a background command. This member is null otherwise.
RESULT	result	Result return value of a command. The result is set when the status reported is SUCCESS, otherwise null is reported.
ERROR	error	Error object describing reason why a command was unsuccessful. Set status is ERROR, otherwise null is reported.

2.2.2 class RESULT

Type	Name	Description
List<DEVICE_OBJECT>	devices	An entry in the devices array if the operation was fetched successfully
List<DEVICE_ERROR>	error	Indicates devices that command failed for, as well as the reason for failure.
List<RS232_DATA_OBJECT>	rs232	List of RS-232 data objects.
List<IR_DATA_OBJECT>	infrared	List of infrared data objects.
List<I2C_EVENT_OBJECT>	i2c_event	List of I2C event objects.
List<CEC_DATA_OBJECT>	cec	List of CEC data objects
List<REBOOT_OBJECT>	reboot	A list of objects that represent the devices for which the request succeeded.
List<SEND_CEC_OBJECT>	send_cec	A list of objects representing devices for which request succeeded.
List<SEND_INFRARED_OBJECT>	send_infrared	A list of objects representing devices for which the request succeeded.
List<SEND_RS232_OBJECT>	send_rs232	A list of objects that represent the devices for which the request succeeded.
List<FIRMWARE_ITEM_OBJECT>	firmware	A list of objects that represent the firmware file.

List<UPDATE_PROGRESS_OBJECT>	update_progress	List of firmware update progress objects.
List<Multicast>	multicast	An array of objects that each represent a multicast IP address.
LAYOUT_DESCRIPTION	layout_description	A layout description object.

2.2.3 class ERROR

Type	Name	Description
string	reason	A machine-readable strings identifying the error reported: AUTHENTICATION: Operation failed because the client was not authenticated. Note: TLS client certificate is currently the only supported means of authentication. ILLEGAL_ARGUMENT**: Invalid argument specified for command. INVALID_STATE**: Operation cannot be performed on the device in its current state. As an example, currently a firmware update is in progress. Tip: Same command may succeed later if condition resolved. INVALID_COMMAND**: Entered command does not exist. Verify the appropriate API module loaded. INVALID_STATE: Operation cannot be performed on device in its current state. For example a firmware update is in progress. Tip: Same command may succeed later if condition resolved. IO_ERROR: Operation failed due to I/O error (disk read/write failure). OUT_OF_MEMORY: Operation failed as no new memory could be allocated. PROTOCOL: Specific to REST API requests. Request failed due to an HTTP protocol-related error (resource not found, bad request, etc.). The HTTP status provides more specific information about the error. UNSUPPORTED_FEATURE: Requested feature or module not supported by BlueRiver API, refer to require command. VERSION: Application not supported by the loaded API module, refer to require command. In addition to the above, the following error responses indicate incorrect behavior on the part of the BlueRiver API (contact BlueRiver support): ● INTERNAL_ERROR ● NOT_IMPLEMENTED

string	message	A error message that is readable by a human, describing the error.
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2.2.4 class DEVICE_OBJECT

Type	Name	Description
string	device_id	Device ID
DEVICE_IDENTITY	identity	Provides information regarding the identity of the device.
DEVICE_CAPABILITIES	capabilities	Provides information regarding the capabilities of the device.
DEVICE_CONFIG	configuration	Provides information regarding the configurable settings of the device.
DEVICE_STATUS	status	Provides information regarding the current status of the device.
List<STREAM_OBJECT>	streams	Describes details regarding the streams of data being sent on the network.
List<SUBSCRIPTION_OBJECT>	subscriptions	Describes details regarding the streams of data being received from the network.
List<NODE>	nodes	Describes device's hardware modules including their status and configuration.

2.2.5 class DEVICE_ERROR

Type	Name	Description
string	device_id	The device ID of the device to which the error belongs.
String	reason	In addition to the errors outlined in Table 2.2.3: BlueRiver error objects, the following errors can occur in response to a device error occurring. These errors are also machine-readable strings identifying error. DEVICE_DISCONNECTED**: Device disconnected from network. DEVICE_TYPE: Operation cannot be performed on device because it does not support it. INVALID_OPERATION: Operation cannot be performed on device for a reason other than device type or state. INVALID_STATE**: Operation cannot be performed on device in its current state. For example a firmware update is in progress. IO_ERROR: Operation failed due to I/O error (e.g. disk read/write failure). REQUEST_TIMEOUT**: Device did not respond in time. RESOURCE_NOT_FOUND: Resource needed for operation not found (e.g. firmware file).
String	message	A error message that is readable by a human, describing the error.

2.2.6 class DEVICE_IDENTITY

Type	Name	Description
string	chipset_type	Identifies the type of the device chipset. Current chipsets are available include NT, NT2000, NT1000, AVP2000T, AVP2000, AVP1000, AVX200T and AVX100.
String	engine	Identifies the type of processing engine the device chipset uses: • BLUERIVER_NT for BlueRiver NT chipset. • PLETHORA for all other chipset types.
Int	vendor_id	Device vendor ID. This feature requires firmware support. If such support is not available, the reserved value 0 is reported.
Int	product_id	Device product ID. This feature requires firmware support. If such support is not available, the reserved value 0 is reported.
String	firmware_comment	Comment string embedded in firmware file that was programmed in the device. This comment typically contains a version string but the exact format and content is device manufacturer-specific. May contain an empty string if no comment is set.
String	firmware_version	Device firmware version.
String	firmware_rc	Device firmware release candidate. Note: For Semtech AptoVision Product Group internal use only.
Bool	is_receiver	True is the device is a receiver device.
Bool	is_transmitter	True is the device is a transmitter device.
List<FIRMWARE_DETAILS>	firmware_details	Provide information regarding the various firmware images of the device.
Int?	chip_id	

2.2.7 class FIRMWARE_DETAILS

Type	Name	Description
string	type	Type of firmware image being described.

		One of the following: PRIMARY for the primary firmware image. GOLDEN for the golden firmware image.
bool	details_available	Whether information is available for the firmware image.
string	firmware_version	Version of the firmware image. If the value of the details_available member is false, this value is undefined.
string	firmware_rc	Release candidate number of the firmware image. If the value of the details_available member is false, this value is undefined. Note: BlueRiver support internal use only.

2.2.8 class DEVICE_CAPABILITIES

Type	Name	Description
bool	color_generator_force	Whether receiver without video processing capability has the ability to blank its video output.
bool	frame_rate_divider	Whether device supports dividing the frame rate by a factor of 2.
bool	hdmi_audio_downmix	Whether device produces a downmix of its HDMI audio. Affects: - Whether device can loop out the HDMI audio downmix to one of its local audio outputs. - Whether a receiver subscribed to this device's HDMI_AUDIO stream can output the HDMI audio downmix to one of its audio outputs.
int	hdmi_inputs	Number of HDMI video inputs.
int	hdmi_outputs	Number of HDMI video outputs.
bool	hdmi_stream1_auto_stop	Whether device supports the HDMI stream 1 auto stop feature.
int	infrared_inputs	Number of infrared receiver connectors.
int	infrared_outputs	Number of infrared emitter connectors.
bool	local_loopback	Whether device supports subscribing to its own video and audio streams.
int	multich_audio_inputs	Number of multichannel audio inputs.
int	multich_audio_outputs	Number of multichannel audio outputs.
bool	nbaset	Whether NBase-T is supported by the device.
bool	net10gbe_dual_link	Whether this is a dual link 10GbE copper+fiber device.

bool	overlay	Whether the monochrome bitmap overlay is supported.
bool	scaler	Supports scaling the video from the HDMI input and sending the scaled video on a separate stream (HDMI stream 1).
int	stereo_audio_inputs	Number of stereo audio inputs.
int	stereo_audio_input_outputs	Number of stereo audio input/outputs.
int	stereo_audio_outputs	Number of stereo audio outputs.
bool	thumbnail	Whether the device supports sending a thumbnail stream.
int	uarts	Number of enabled UARTs.
bool	usb_hid	Whether USB HID is supported.
bool	usb_icron	Whether device has an Icron USB extender.
bool	video_processing	Whether the receiver supports video processing. This includes support for video processing display modes, such as fast switch, multiview and wall.

2.2.9 class DEVICE_CONFIG

Type	Name	Description
string	device_name	Name of the device.
bool	locate_mode	Whether the device is in locate mode. TIP: When in locate mode, all device LEDs flash so it can be located easily.
string	secure_av	secure av "default" or "user_defined".
bool	resume_streaming	Indicates whether a device is to automatically restart streams and subscriptions after a reboot.
bool	claimed	Indicates if device has been exclusively claimed by a BlueRiver Control Server (API).
bool	authenticated	TRUE indicates device is authenticated bound to an API server. FALSE indicates the device is not.
string	device_mode	Indicates whether the configured device mode is TRANSMITTER, RECEIVER or TRANSCEIVER.

2.2.10 class DEVICE_STATUS

Type	Name	Description
bool	active	Indicates if device is connected to the network.
bool	point_to_point	Indicates if device is in point-to-point mode.
string	point_to_point_peer	Device ID of the peer device. Is set to NULL when point_to_point is

		false or the peer device does not support this feature.
string	boot_status	Firmware image device booted with. One of the following: - PRIMARY for the primary image - GOLDEN for the golden (fallback) image - UNKNOWN if active firmware cannot be determined because device firmware does not support this feature. If the value reported is GOLDEN, this indicates that the primary firmware image is corrupted and device may need to be re-programmed.
bool	update_in_progress	Indicates if a firmware update or first-time programming is running on the device. Once the firmware programming procedure is complete, the device must be rebooted, at which point the value of this member is reset to false.
string	igmp_reporting_mode	Indicates whether the IGMP reporting mode is INTERVAL (sent unsolicited every minute) or QUERY (sent as a response to the IGMP snooping queries from the switch).
DEVICE_ERROR_STATUS	error_status	Indicates if the device is reporting error code.
int	temperature	Current chipset junction temperature in degrees Celsius.

2.2.11 class DEVICE_ERROR_STATUS

Type	Name	Description
bool	has_error_code	Indicates if the device is reporting an error code.
int	error_code	If has_error_code member is true, this indicates that the member contains an error code reported by the device. Otherwise, the value of this member is undefined. Reminder: Meaning of error code device-dependent.

2.2.12 class STREAM_OBJECT

Type	Name	Description
string	type	Type of data contained in the stream. STEREO_AUDIO: Audio from an single I2S/analog source

		CEC: CEC data HDMI: HDMI video HDMI_AUDIO: HDMI audio MULTICH_AUDIO: I2S audio INFRARED: Infrared remote-control data RS232: RS-232 data
int	index	Index of the stream of the given type (streams are indexed individually by type, not necessarily contiguous)
STREAM_CONFIG	configuration	Stream configuration
STREAM_STATUS	status	STREAMING: Subscription is running STOPPED: Subscription is stopped LINK_DOWN: The 10-Gigabit network interface is down
SOURCE_REFERENCE	source	Identifies the origin of the streamed data.

2.2.13 class STREAM_CONFIG

Type	Name	Description
string	address	IP address that the data is to be sent to.
int?	dest_index	
bool	enable	Whether the stream is enabled or not.
bool	auto_stop	When <code>true</code> , the stream stops automatically whenever the video source is lost or a resolution change occurs. Option useful when managing network bandwidth management. A video resolution change leads to a change in bandwidth usage by the video. It could be appropriate to take an action in response before starting the stream again. For example, enable or disable compression. Always <code>false</code> for streams other than the HDMI scaled stream (HDMI : 1).
bool	remove_audio	When <code>true</code> , HDMI audio is not sent with the stream. Reminder: Audio is only present on the first HDMI stream, that is HDMI 0.
SOURCE_SELECTION	source	Selection of the input source (optional). This member is only present if the input source is configurable on the specified stream.

2.2.14 class STREAM_STATUS

Type	Name	Description
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string	state	State of the subscription: STREAMING: Subscription is running STOPPED: Subscription is stopped LINK_DOWN: The 10-Gigabit network interface is down
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2.2.15 class SUBSCRIPTION_OBJECT

Type	Name	Description
string	type	Type of data, same values as for stream type. STERO_AUDIO: Audio from an single I2S/analog source CEC: CEC data HDMI: HDMI video HDMI_AUDIO: HDMI audio MULTICH_AUDIO: I2S audio INFRARED: Infrared remote-control data RS232: RS-232 data
int	index	Index of subscription of given type. Note: Subscriptions indexed individually by type, therefore not necessarily contiguous.
SUBSCRIPTION_CONFIG	configuration	Subscription configuration.
SUBSCRIPTION_STATUS	status	Subscription status.

2.2.16 class SUBSCRIPTION_CONFIG

Type	Name	Description
string	address	IP address from which to receive the data.
bool	enable	Whether the subscription is enabled or not
int?	mute_msec	Amount of milliseconds firmware mutes on a HDMI_AUDIO subscription change. Value must be between 0 and 5000 inclusively (present for HDMI_AUDIO subscription only).

2.2.17 class SUBSCRIPTION_STATUS

Type	Name	Description
string	state	State of the subscription: STREAMING: Subscription is running STOPPED: Subscription is stopped LINK_DOWN: The 10-Gigabit network interface is down

2.2.18 class NODE

Type	Name	Description
NODE_TYPE	type	Type of node. STEREO_AUDIO_INPUT STEREO_AUDIO_INPUT_OUTPUT STEREO_AUDIO_OUTPUT BUTTON FRAME_BUFFER FRAME_RATE_CONVERTER GPIO HDMI_DECODER HDMI_ENCODER HDMI_MONITOR MULTICH_AUDIO_INPUT MULTICH_AUDIO_OUTPUT INFRARED_DECODER INFRARED_ENCODER LED NETWORK_INTERFACE NETWORK_SWITCH NULL_SOURCE SCALER UART USB_ICRON VIDEO_INPUT Other types could be present and should be ignored.
int	index	Index of the node of the given type. Nodes are indexed independently for each type, not necessarily contiguous.
List<NODE_INPUT>	inputs	Array describing how the node is connected to upstream nodes and/or subscriptions.

2.2.19 class NODE_INPUT

Type	Name	Description
string	name	Input name, allowable values depend on node type.
int	index	Index of the input with the given name. Inputs are indexed independently for each name.
NODE_INPUT_CONFIG	configuration	Input configuration.

NODE_INPUT_STATUS	status	Input status.
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2.2.20 class NODE_INPUT_CONFIG

Type	Name	Description
SOURCE_SELECTION	source	(Optional) Selection of the input source. Only present if input source is configurable on specified input.

2.2.21 class NODE_INPUT_STATUS

Type	Name	Description
SOURCE_REFERENCE	source	Current input source.

2.2.22 class SOURCE_REFERENCE

Type	Name	Description
string	ref_class	The reference class reported. Is one of the following: NODE: The origin of the data is a node. SUBSCRIPTION: The origin of the data is a subscription. In addition, the following value could be present in the configuration choices of a node input: AUTOMATIC: Source is automatically selected. For example, when a valid signal is present.
string	ref_type	Reference type, matches type member of a node or subscription. Undefined if reference class is AUTOMATIC.
int	ref_index	Reference index, matches index member of a node or subscription. Undefined if reference class is AUTOMATIC.

2.2.23 class SOURCE_SELECTION

Type	Name	Description
int	value	Selected source, must match the value member of one of the enumerated choices.
List<SOURCE_SELECTION_CHOICE>	choices	Enumeration of the allowed source selections.

2.2.24 class SOURCE_SELECTION_CHOICE

Type	Name	Description
int	value	Value of this choice.
string	description	Human-readable description of this choice.
SOURCE_REFERENCE	details	Source reference associated with this choice.

2.2.25 class NODE_STEREO_AUDIO_INPUT : NODE

Type	Name	Description
NODE_TYPE	type	STEREO_AUDIO_INPUT

2.2.26 class NODE_STEREO_AUDIO_INPUT_OUTPUT : NODE

Type	Name	Description
NODE_TYPE	type	STEREO_AUDIO_INPUT_OUTPUT
STEREO_AUDIO_INPUT_OUTPUT_NODE_CONFIG	configuration	

2.2.27 class STEREO_AUDIO_INPUT_OUTPUT_NODE_CONFIG

Type	Name	Description
string	direction	Port direction, which is one of the following: INPUT: The port is set to be an audio input. OUTPUT: The port is set to be an audio output.

2.2.28 class NODE_STEREO_AUDIO_OUTPUT : NODE

Type	Name	Description
NODE_TYPE	type	STEREO_AUDIO_OUTPUT

2.2.29 class NODE_BUTTON : NODE

Type	Name	Description
NODE_TYPE	type	BUTTON
BUTTON_NODE_CONFIG	configuration	

2.2.30 class BUTTON_NODE_CONFIG

Type	Name	Description
BUTTON_ACTION_CONFIG	action	Action triggered when this button is pressed.

2.2.31 class BUTTON_ACTION_CONFIG

Type	Name	Description
int	value	Button action selection, must match the value member of one of the choices listed in the choices member. Note: Not all actions available for all buttons or devices. Valid choices are listed in the choices member.

List<BUTTON_ACTION_CHOICE>	choices	Button action choices.
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2.2.32 class BUTTON_ACTION_CHOICE

Type	Name	Description
int	value	Value of this choice.
string	description	Human-readable description of this choice.
string	details	Machine-readable name of action, will be one of the following: NONE: no action SEND_EDID: send monitor EDID to all transmitter devices. SWITCH_VIDEO_INPUT: switch between video inputs.

2.2.33 class NODE_CEC : NODE

Type	Name	Description
NODE_TYPE	type	CEC
CEC_NODE_CONFIG	configuration	

2.2.34 class CEC_NODE_CONFIG

Type	Name	Description
NODE_TYPE	type	CEC
string	p2p_mode	If AVP device is operating in Point-to-Point (p2p_mode) mode, CEC Node can be configured to operate in API or Passthrough (default) mode. When in API mode it allows CEC data to be injected from API to logical address "14" (0x0E – Free use) and when operating in Passthrough mode, CEC data is passed between paired devices.

2.2.35 class NODE_COLOR_GENERATOR : NODE

Type	Name	Description
NODE_TYPE	type	COLOR_GENERATOR
COLOR_GENERATOR_NODE_CONFIG	configuration	

2.2.36 class COLOR_GENERATOR_NODE_CONFIG

Type	Name	Description
bool	enable	When true, blank video output with a uniform color. Useful to implement video mute functionality. Note: Only supported for decoders (RX) operating in a display mode other than genlock.
string	color	String representation of the color to use for blanking.

		String format is RRGGBB, where RR, GG and BB are each a pair of hexadecimal digits that respectively represent the 8-bit value of the red, green and blue components. These 8-bit values must be interpreted in the context of the default quantization range (full or limited) for the current output video format as specified in standard CEA 861-F. Note: Only supported when decoder (RX) operating in a display mode other than genlock.
bool	force_output	When <code>true</code> , device outputs a black video signal with fixed 720p60 format.

2.2.37 class NODE_FRAME_BUFFER : NODE

Type	Name	Description
NODE_TYPE	type	FRAME_BUFFER
FRAME_BUFFER_NODE_CONFIG	configuration	
FRAME_BUFFER_NODE_STATUS	status	

2.2.38 class FRAME_BUFFER_NODE_CONFIG

Type	Name	Description
string	display_mode	Processing display mode. Will be one of the following: FAST_SWITCHED: Full screen fast switch mode GENLOCK_SCALING: Full screen zero-frame latency scaling mode WALL_FAST_SWITCHED: Fast switch display wall mode WALL_GENLOCKED: Zero-frame latency display wall mode MULTIVIEW: Multiview mode
int	width	Video width
int	height	Video height
int	horiz_offset	Video horizontal offset
int	vert_offset	Video vertical offset
int	frames_per_second	Frame rate
int	keep_width	Width of wall region of interest, in source pixels.
int	keep_height	Height of wall region of interest, in source pixels.
int	viewport_horiz	Viewport horizontal position. Only relevant in wall display mode (fast switched or genlocked wall).
int	viewport_vert	Viewport vertical position. Only relevant in wall display mode (fast

		switched or genlocked wall).
int	viewport_width	Viewport width. Only relevant in wall display mode (fast switched or genlocked wall).
int	viewport_height	Viewport width. Only relevant in wall display mode (fast switched or genlocked wall).
string	aspect_ratio_mode	The aspect ratio applied to the output. Options are <code>KEEP</code> , <code>CROP</code> and <code>STRETCH</code> .
bool	has_auto_wall_details	Indicates if the arguments for the Source Independent (auto_wall) are currently applied. TRUE indicates this feature enabled and FALSE that it is not.

2.2.39 class FRAME_BUFFER_NODE_STATUS

Type	Name	Description
VIDEO_FORMAT	source_video	Format of video source (full screen and wall modes only)

2.2.40 class VIDEO_FORMAT

Type	Name	Description
int	width	Video width
int	height	Video height
int	frames_per_second	Frame rate. Note: The value of this member is always an integer. This means for example, 59.94 fps is reported as 60. For more precision, consider using the frame_rate member of the Video_Format_Details object instead.
string	color_space	Color space, which is one of the following: RGB: RGB graphics YCBCR_444: YcbCr 4:4:4 video YCBCR_422: YcbCr 4:2:2 video YCBCR_420: YcbCr 4:2:0 video
int	bits_per_pixel	Number of bits per video component, one of: 8-bits 10-bits 12-bits
string	scan_mode	Video scan mode, which is one of the following: PROGRESSIVE: Returned for progressive video INTERLACED: Returned for interlaced video

2.2.41 class NODE_FRAME_RATE_CONVERTER : NODE

Type	Name	Description
NODE_TYPE	type	FRAME_RATE_CONVERTER
FRAME_RATE_CONVERTER_NODE_STATUS	status	

2.2.42 class FRAME_RATE_CONVERTER_NODE_STATUS

Type	Name	Description
int	divide_factor	Factor by which the frame rate is divided.

2.2.43 class NODE_GPIO : NODE

Type	Name	Description
NODE_TYPE	type	GPIO
GPIO_NODE_CONFIG	configuration	GPIO node configuration detail
GPIO_NODE_STATUS	status	GPIO node configuration Status

2.2.44 class GPIO_NODE_CONFIG

Type	Name	Description
int	tristate	Current direction of the GPIO: 0 -- indicates the pin is an input. 1 -- indicates the pin is an output.
int	cfg_val	The output value (0 or 1) currently configured for the GPIO pin.

2.2.45 class GPIO_NODE_STATUS

Type	Name	Description
int	stat_val	The value (0 or 1) read from the GPIO pin.

2.2.46 class NODE_HDMI_DECODER : NODE

Type	Name	Description
NODE_TYPE	type	HDMI_DECODER
HDMI_DECODER_NODE_CONFIG	configuration	HDMI Decoder node configuration detail
HDMI_DECODER_NODE_STATUS	status	HDMI Decoder node configuration Status

2.2.47 class HDMI_DECODER_NODE_CONFIG

Type	Name	Description
string	edid	Configured EDID seen by connected HDMI sources. Format of member is hexadecimal string with 512-character length representing 256 bytes of data.
bool	hdcp_support_enable	Enable HDCP support.

		When disabled, the device reports to the HDMI source that it does not support HDCP. TIPS: ◆ Some laptops and possibly other HDMI sources always encrypt video with HDCP if the sink connected to it supports it, even when not playing back HDCP protected content. ◆ Can cause problems when trying to display video from HDMI sources on monitors without HDCP support. ◆ Disabling HDCP support in the transmitter device solves this issue.
bool	hdcp_22_support_disable	Disable HDCP 2.2 support. When set <code>true</code>, device reports to HDMI source it only supports HDCP 1.4 (if <code>hdcp_support_enable</code> is also <code>true</code>).

2.2.48 class HDMI_DECODER_NODE_STATUS

Type	Name	Description
bool	hdcp_protected	Indicates if HDMI source is HDCP protected.
string	hdcp_version	Version of HDCP content protection: NONE: Content is not HDCP-protected. HDCP_1_4: Supports HDCP 2.1 and below (including 1.4). HDCP_2_2: HDCP 2.2.
bool	hdmir20support	Indicates if HDMI decoder supports HDMI 2.0 resolutions, such as 4K60.
bool	source_stable	Indicates if device is receiving video from a connected source.
VIDEO_FORMAT	Video	Indicates format of video received from the source, only valid if <code>source_stable</code> is <code>true</code>. Undefined if <code>source_stable</code> is <code>false</code>.
bool	has_video_details	Indicates if detailed video format and timing information available from device.

VIDEO_FORMAT_DETAILS	video_details	Detailed video format and timing information. Undefined when <code>source_stable</code> or <code>has_video_details</code> is false.
bool	has_audio_details	Indicates whether audio format information available from device.
AUDIO_FORMAT_DETAILS	audio_details	Information regarding audio format. Undefined if <code>source_stable</code> or <code>has_audio_details</code> is false.
bool	has_audio_details_extended	Indicates if extended audio format information available from the device.
AUDIO_FORMAT_EXTENDED	audio_details_extended	Extended data regarding audio format. Undefined if either <code>source_stable</code> or <code>has_audio_details_extended</code> is false.
bool	audio_downmix_support	Indicates if HDMI decoder generates a downmix version of the extracted HDMI audio.

2.2.49 class VIDEO_FORMAT_DETAILS

Type	Name	Description
double	frame_rate	Frame rate, includes fractional values (e.g. 59.54).
int	pixel_clock	Pixel clock, in hertz (Hz).
int	total_width	Total horizontal resolution including blanking.
int	total_height	Total vertical resolution including blanking.
int	hsync_width	Horizontal sync pulse width in pixels.
int	hsync_front_porch	Horizontal sync front porch in pixels.
bool	hsync_negative	Indicates whether horizontal sync has negative polarity.
int	vsync_width	Vertical sync pulse width, in pixels.
int	vsync_front_porch	Vertical sync front porch, in pixels.
bool	vsync_negative	Indicates whether vertical sync has negative polarity.
int	vic	AVI InfoFrame Video Information Code (VIC).

bool	has_hdmi_vic	Indicates whether there is an HDMI Vendor-Specific InfoFrame with an HDMI_VIC.
int	hdmi_vic	HDMI Vendor-Specific Infoframe HDMI_VIC. Undefined if has_hdmi_vic is false.
string	picture_aspect	AVI InfoFrame picture aspect ratio: ASPECT_NO_DATA: for No Data ASPECT_4_3: for a 4:3 aspect ratio ASPECT_16_9: for a 16:9 aspect ratio
bool	has_active_format	Indicates whether the AVI InfoFrame specifies active format information.
int	active_format	AVI InfoFrame Active Format Description (AFD) code. For more information refer to the CEA-861-F Annex H.
string	it_content_type	AVI InfoFrame IT content type: NO_DATA: for No Data GRAPHICS: for Graphics type PHOTO: for Photo type CINEMA: for Cinema type GAME: for Game type
string	scan_information	AVI InfoFrame scan information: NO_DATA: for No Data OVERSCAN: for overscan UNDERSCAN: for underscan
string	colorimetry	AVI InfoFrame colorimetry: NO_DATA for No Data RGB for RGB BT601 for SMPTE 170M/ITU-R BT.601 BT709 for ITU-R BT.709 XVYCC601 for xvYCC ₆₀₁ XVYCC709 for xvYCC ₇₀₉ SYCC601 for sYCC ₆₀₁ ADOBE_YCC601 for Adobe _{YCC601} ADOBE_RGB for Adobe _{RGB} BT2020_YCC_CONST for ITU-R BT.2020 Y'C _B C' _{RC} BT2020_YCC for ITU-R BT.2020 Y'C _B C' _R BT2020_RGB for ITU-R BT.2020 R'G'B' This member combines information from the “RGB or YC _B C _R ” (Y), Colorimetry (C) and Extended Colorimetry (EC) fields of the AVI InfoFrame. See CEA-861-F Table 13.

string	rgb_range	AVI InfoFrame RGB quantization range: DEFAULT LIMITED FULL
string	ycc_range	AVI InfoFrame YCbCr quantization range: LIMITED FULL
bool	has_hdr	Indicates whether High-Dynamic Range video format information is available from the device.
string	hdr	High-Dynamic Range video format. Can be one of the following: SDR Standard dynamic range DV_LLAT_RGB10_12 Dolby Vision Low Latency mode 10/12-bit RGB 4:4:4 DV_LLAT_YCBCR10_12_444 Dolby Vision Low Latency mode 10/12-bit YCbCr 4:4:4 DV_LLAT_YCBCR12_422 Dolby Vision Low Latency mode 12-bit YCbCr 4:2:2 DV_STD_RGB8 Dolby Vision Standard mode Tunnel 8-bit RGB 4:4:4 DV_STD_YCBCR12 Dolby Vision Standard mode Native 12-bit YCbCr 4:2:2 HDR10 HDR10 - SMPTE ST 2084 HLG Hybrid Log Gamma – ITU-R BT.2100-0 This member is undefined if has_hdr member is reported as false.

2.2.50 class AUDIO_FORMAT_DETAILS

Type	Name	Description
int	sampling_frequency	The sampling frequency, in hertz (Hz).
int	number_of_channels	The number of audio channels

2.2.51 class AUDIO_FORMAT_EXTENDED

Type	Name	Description
int?	channel_allocation	The channel allocation as reported in the CEA-861-F audio InfoFrame. Note: Valid values and their meanings are defined in CEA-861-F Table 31.

int?	sample_size	The size of each audio sample in bits. An integer value between 16 and 24 inclusive or zero (0) if it is undefined.
int?	level_shift_value	Level Shift Value as reported in the CEA-861-F audio InfoFrame. Provides information on the downmix coefficients attenuation. Note: Valid values and their meanings are defined in CEA-861-F Table 32.
int?	lfe_playback_level	The Low-Frequency Effect (LFE) playback level as reported in the CEA-861-F audio InfoFrame. Note: Refer to CEA-861-F Table 34 for valid values and their meanings.
string?	audio_encoding_type	Specifies the type of audio encoding. LPCM Linear PCM audio. ENCODED_HBR High-Bitrate audio. ENCODER_OTHER Other type of audio, such as compressed audio.
bool?	is_downmix_inhibited	Downmix inhibit as reported in the CEA-861-F audio InfoFrame. If true, audio output is prohibited from being downmixed. Otherwise, either downmixing is permitted or no information is available.

2.2.52 class NODE_HDMI_ENCODER : NODE

Type	Name	Description
NODE_TYPE	type	HDMI_ENCODER
HDMI_ENCODER_NODE_CONFIG	configuration	HDMI Encoder node configuration detail
HDMI_ENCODER_NODE_STATUS	status	HDMI Encoder node configuration Status

2.2.53 class HDMI_ENCODER_NODE_CONFIG

Type	Name	Description
string	hdcp_output_mode	The following nodes supported for the HDCP Output mode: UNSUPPORTED ● The feature is unsupported on this device FOLLOW_SINK_1 ● Sets HDCP output level to the highest HDCP capability of sink. FOLLOW_SINK_2 ● Sets HDCP output level to HDCP 1.4 when possible. FOLLOW_SOURCE ● Sets HDCP output level to same as the source.

2.2.54 class HDMI_ENCODER_NODE_STATUS

Type	Name	Description
bool	hdcp_protected	Indicates if HDCP protection is enabled. Automatically enabled if source is HDCP protected.
bool	hdmi_2_0_support	Indicates if the HDMI encoder supports HDMI 2.0 resolutions (e.g. 4K60).
bool	source_stable	Indicates device is receiving video from the source.
VIDEO_FORMAT	Video	Format of video being output by the HDMI encoder. Undefined if <code>source_stable</code> is false.
bool	has_video_details	Indicates whether detailed video format and timing information is available from this device.
VIDEO_FORMAT_DETAILS	video_details	Detailed video format and timing information. Undefined if <code>source_stable</code> or <code>has_video_details</code> is false.

2.2.55 class NODE_HDMI_MONITOR : NODE

Type	Name	Description
NODE_TYPE	type	HDMI_MONITOR
HDMI_MONITOR_NODE_STATUS	status	HDMI Monitor node status

2.2.56 class HDMI_MONITOR_NODE_STATUS

Type	Name	Description
bool	connected	Indicates if there is a connected monitor (HPD – Hot-Plug Detect).
string	edid	EDID of the connected monitor. Format of this member is a hexadecimal string with a length of 512 characters (represents 256 bytes of data). If no monitor is connected (see <code>connected</code> member), member contains an empty string.

2.2.57 class NODE_MULTICH_AUDIO_INPUT : NODE

Type	Name	Description
NODE_TYPE	type	MULTICH_AUDIO_INPUT
MULTICH_AUDIO_INPUT_NODE_CONFIG	configuration	Multichannel Audio input node configuration

2.2.58 class MULTICH_AUDIO_INPUT_NODE_CONFIG

Type	Name	Description
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SAMPLING_FREQUENCY_CONFIG	sampling_frequency	Audio sampling rate configuration
CHANNELS_NUMBER_CONFIG	number_of_channels	Number of audio channels configuration
CHANNELS_ALLOCATION_CONFIG	channel_allocation	HDMI audio channel allocation as reported in CEA-861-F audio infoframe, from range of 0 to 0x31.
string	audio_encoding_type	Audio encoding type. Choices are LPCM, ENCODED_HBR or ENCODED_OTHER

2.2.59 class SAMPLING_FREQUENCY_CONFIG

Type	Name	Description
int	value	Sampling frequency selection. This value is sampling frequency in Hertz and must match value member of an option listed in the choices member. Note: Not all sampling rates are supported by all devices. The valid choices are listed in the choices member.
List<SAMPLING_FREQUENCY_CHOICE>	choices	Sampling rate choices.

2.2.60 class SAMPLING_FREQUENCY_CHOICE

Type	Name	Description
int	value	Sampling frequency in Hertz.
string	description	Human-readable description of this choice.

2.2.61 class CHANNELS_NUMBER_CONFIG

Type	Name	Description
int	value	Number of channels selection. Values from 2 to 8 represent the number of audio channels. Other values are reserved for future use. This value must match the value member of one of the choices listed in the choices member. Note: Not all number of channels are supported by all devices. The valid choices are the ones listed in the choices member.
List<CHANNELS_NUMBER_CHOICE>	choices	Number of audio channels choices.

2.2.62 class CHANNELS_NUMBER_CHOICE

Type	Name	Description
int	value	Value of this choice. Values from 2 to 8 represent the number of audio channels. Other values are reserved for future use.
string	description	Human-readable description of this choice.

2.2.63 class CHANNELS_ALLOCATION_CONFIG

Type	Name	Description
int	value	Channel allocation selection. Values from 0 to 0x31 represent the audio channel allocation of the incoming multi-channel audio. The channel allocation value is as reported in the CEA-861-F documentation. Other values are reserved for future use. This value must match the value member of one of the choices listed in the choices member. Note: Not all channel allocations are supported by all devices. The valid choices are listed in the choices member.
List<CHANNEL_ALLOCATION_CHOICE>	choices	Channel Allocation choices.

2.2.64 class CHANNEL_ALLOCATION_CHOICE

Type	Name	Description
int	value	Value of this choice. Values from 0 to 0x31 represent the audio channel allocation configuration. Other values are reserved for future use.
string	description	Human-readable description of this choice.

2.2.65 class NODE_MULTICH_AUDIO_OUTPUT : NODE

Type	Name	Description
NODE_TYPE	type	MULTICH_AUDIO_OUTPUT

2.2.66 class NODE_INFRARED_DECODER : NODE

Type	Name	Description
NODE_TYPE	type	INFRARED_DECODER

2.2.67 class NODE_INFRARED_ENCODER : NODE

Type	Name	Description
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NODE_TYPE	type	INFRARED_ENCODER
INFRARED_ENCODER_NODE_STATUS	status	Infrared encoder node status

2.2.68 class INFRARED_ENCODER_NODE_STATUS

Type	Name	Description
int	data_length_max	Maximum length, in bytes, of the Pronto code that can be sent to the infrared encoder.

2.2.69 class NODE_LED : NODE

Type	Name	Description
NODE_TYPE	type	LED
LED_NODE_CONFIG	configuration	LED node configuration detail

2.2.70 class LED_NODE_CONFIG

Type	Name	Description
LED_FUNCTION_CONFIG	function	LED Function

2.2.71 class LED_FUNCTION_CONFIG

Type	Name	Description
int	value	LED function selection. When setting this property using the set property command, the following values are allowed: 1 -- blinking LED 2 -- power LED 3 -- video is stable indicator 4 -- network TX indicator 5 -- network RX indicator 6 -- always ON 7 -- always OFF 8 -- default function for this LED Device firmware allows LED function to be set but does not report current function back to the API. For this reason, value reported in this member always: 0 -- current function is unknown Note: Not all functions are available for all LEDs. Valid choices are listed in the choices member.
List<LED_FUNCTION_CHOICE>	choices	LED function choices

2.2.72 class LED_FUNCTION_CHOICE

Type	Name	Description
int	value	Value of this choice
string	description	Human-readable description of this choice.
string	details	Machine-readable name of this function. Is one of the following: UNKNOWN Current function is unknown BLINK Blinking LED POWER Power LED VIDEO_STABLE Video is stable NETWORK_TX Network TX indicator NETWORK_RX Network RX indicator ALWAYS_ON Always ON ALWAYS_OFF Always OFF DEFAULT Default function for this LED

2.2.73 class NODE_NETWORK_INTERFACE : NODE

Type	Name	Description
NODE_TYPE	type	NETWORK_INTERFACE
NETWORK_INTERFACE_NODE_CONFIG	configuration	Network Interface node configuration detail
NETWORK_INTERFACE_NODE_STATUS	status	Network Interface node configuration status

2.2.74 class NETWORK_INTERFACE_NODE_CONFIG

Type	Name	Description
string	hostname	Host name for the device on this interface
NET_INTERFACE_IP_CONFIG	ip	Internet Protocol (IP) configuration

2.2.75 class NET_INTERFACE_IP_CONFIG

Type	Name	Description
string	address	IP address.
string	mode	One of the following: MANUAL Internet Protocol configuration is fixed and configured manually. DHCP Internet Protocol configuration is obtained from a DHCP server, with fallback on auto-IP if no DHCP server is available.
string	mask	IP subnet mask

string	gateway	IP address of default gateway
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2.2.76 class NETWORK_INTERFACE_NODE_STATUS

Type	Name	Description
string	mac_address	MAC address for the interface.
NET_INTERFACE_IP_STATUS	ip	Internet Protocol (IP) status.

2.2.77 class NET_INTERFACE_IP_STATUS

Type	Name	Description
string	address	IP address

2.2.78 class NODE_NETWORK_PORT : NODE

Type	Name	Description
NODE_TYPE	type	NETWORK_PORT
NETWORK_PORT_NODE_CONFIG	configuration	Network Port node configuration detail
NETWORK_PORT_NODE_STATUS	status	Network Port node configuration status

2.2.79 class NETWORK_PORT_NODE_CONFIG

Type	Name	Description
string	Speed	Speed configuration for the network port. Is one of the following: SPEED_10G fixed 10Gbps speed Note: Supported only if supports_speed_10g status member of this node is true. SPEED_5G fixed 5Gbps speed Note: Supported only if the supports_speed_5g status member of this node is true. SPEED_2_5G fixed 2.5Gbps speed Note: Supported only if the supports_speed_2_5g status member of this node is true. AUTO_10G auto-negotiated up to 10Gbps speed Note: Supported only if the supports_auto_10g status member of this node is true. AUTO_5G auto-negotiated up to 10Gbps speed Note: Supported only if the supports_auto_10g status member of this node is true.

2.2.80 class NETWORK_PORT_NODE_STATUS

Type	Name	Description
bool	active	Indicates if the port currently selected as active. On dual-10GbE port devices with automatic switchover, only one port is active at a time.
double	Speed	The current speed of the port in Gbps.
bool	supports_speed_10g	Indicates if the <code>SPEED_10G</code> value is supported for the speed configuration member.
bool	supports_speed_5g	Indicates if the <code>SPEED_5G</code> value is supported for the speed configuration member.
bool	supports_speed_2_5g	Indicates if the <code>SPEED_2_5G</code> value is supported for the speed configuration member.
bool	supports_auto_10g	Indicates if the <code>AUTO_10G</code> value is supported for the speed configuration member.
bool	supports_auto_5g	Indicates if the <code>AUTO_5G</code> value is supported for the speed configuration member.

2.2.81 class NODE_NETWORK_SWITCH : NODE

Type	Name	Description
NODE_TYPE	type	NETWORK_SWITCH
NETWORK_SWITCH_NODE_CONFIG	configuration	Network Switch node configuration detail

2.2.82 class NETWORK_SWITCH_NODE_CONFIG

Type	Name	Description
bool	gigabit_port_1_enable	Enable first Gigabit Ethernet port on the device

2.2.83 class NODE_NULL_SOURCE : NODE

Type	Name	Description
NODE_TYPE	type	NULL_SOURCE

2.2.84 class NODE_SCALER : NODE

Type	Name	Description
NODE_TYPE	type	SCALER
SCALER_NODE_CONFIG	configuration	Scaler node configuration detail
SCALER_NODE_STATUS	status	Scaler node configuration status

2.2.85 class SCALER_NODE_CONFIG

Type	Name	Description
int	width	Scaled video width, in pixels
int	height	Scaled video height, in pixels

2.2.86 class SCALER_NODE_STATUS

Type	Name	Description
VIDEO_FORMAT	video	Video format detail

2.2.87 class NODE_UART : NODE

Type	Name	Description
NODE_TYPE	type	UART
UART_NODE_CONFIG	configuration	UART node configuration detail
UART_NODE_STATUS	status	UART node configuration status

2.2.88 class UART_NODE_CONFIG

Type	Name	Description
int	baud_rate	Baud rate.
int	data_bits	Number of data bits: 6, 7 or 8.
int	stop_bits	Number of stop bits: 1 or 2.
string	parity	Parity: NONE, ODD or EVEN.

2.2.89 class UART_NODE_STATUS

Type	Name	Description
string	usage	Indicates the UART port usage. Can have one of the following values: EXPOSED Port with an external connector RESERVED Port internally connected, for example to an external MCU.

2.2.90 class NODE_USB_HID : NODE

Type	Name	Description
NODE_TYPE	type	USB_HID
USB_HID_NODE_CONFIG	configuration	USB HID node configuration detail
USB_HID_NODE_STATUS	status	USB HID node configuration status

2.2.91 class USB_HID_NODE_CONFIG

Type	Name	Description
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string	role	The role of the device as a USB extender: <code>DISABLED</code> USB support is disabled. <code>LOCAL</code> for a USB device that connects to a host (e.g. PC). <code>REMOTE</code> for a USB host to which devices connect (keyboard, mouse, etc.).
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2.2.92 class USB_HID_NODE_STATUS

Type	Name	Description
List<USB_DEVICE_STATUS>	usb_devices	The status of USB devices. ● If device configured as a USB host (<code>REMOTE</code> role), member reports status of USB devices connected to it. ● If device configured as a USB device (<code>LOCAL</code> role), reports status of USB devices on the device to which it is paired. ● If USB support disabled, content of this member unspecified.
USB_HOST_STATUS	usb_host	Status of the USB host (e.g. PC) connected to this device. Member must be ignored if device configured as other than a USB device (i.e. USB host or disabled).

2.2.93 class USB_DEVICE_STATUS

Type	Name	Description
string	status	One of the following: <code>DISCONNECTED</code> No USB device is connected. <code>CONFIGURING</code> USB device being configured following device connection. This is a transient state. <code>UNSUPPORTED</code> Connected USB device not supported. Example, not a USB HID device. <code>CONNECTED</code> USB device connected, configured and ready to use.

2.2.94 class USB_HOST_STATUS

Type	Name	Description
string	status	One of the following: <code>DISCONNECTED</code> Device is disconnected from the host. <code>CONFIGURING</code> Device being configured by the USB host following connection. This is a transient state. <code>CONNECTED</code> Device is connected to USB host and fully configured.

2.2.95 class NODE_USB_ICRON : NODE

Type	Name	Description
NODE_TYPE	type	USB_ICRON
USB_ICRON_CHIP_NODE_CONFIG	configuration	USB Icron Chip node configuration detail
USB_ICRON_CHIP_NODE_STATUS	status	USB Icron Chip node configuration status

2.2.96 class USB_ICRON_CHIP_NODE_CONFIG

Type	Name	Description
string	extender_type	The type of extender: LOCAL for a local extender (LEX). REMOTE for a remote extender (REX).
string	program_mode	Currently enabled programming mode for the USB extender. One of the following: FIRMWARE for the extender's firmware. EEPROM for the extender's non-volatile configuration memory. NONE if programming is disabled
int	rs232_port	The (zero-based) number of the RS-232 port through which programming of the USB extender is enabled. If value of the program_mode member is NONE, value is undefined.

2.2.97 class USB_ICRON_CHIP_NODE_STATUS

Type	Name	Description
bool	chip_present	Indicates if an Icron chip is present. Property is false if device board has a connector for an Icron USB extender daughter board but one is not installed.
string	mac_address	MAC address of the ICRON chip
bool	reachable	Indicates that the USB extender network interface is reachable (TRUE) or not (FALSE) with an IP address.
string	revision	The revision level of the Icron firmware.
string	ip_mode	The active IP mode, either DHCP or manual (static).
string	ip_address	
string	peer_mac_address	If paired, provides MAC address of the peer extender. If the extender is not paired or an Icron module has not been detected then this field is blank ("").

string	peer_device_id	If currently paired, the device ID of the paired AVP chipset is shown that is associated with the peer Icron device. If the Icron extender is not paired or an Icron module has not been detected then this field is blank ("").
bool	link_status_support	Indicates whether the Icron supports the linked attribute.
bool	linked	Indicates if bidirectional communication is currently established. Devices need to negotiate the link before they can successfully link. TRUE indicates this negotiation successful. If devices currently are not paired or this link is not successfully established, the status is FALSE. The field is also FALSE if the link_status_support field reports FALSE.

2.2.98 class NODE_VIDEO_INPUT : NODE

Type	Name	Description
NODE_TYPE	type	VIDEO_INPUT
VIDEO_INPUT_NODE_STATUS	status	Video input node configuration status

2.2.99 class VIDEO_INPUT_NODE_STATUS

Type	Name	Description
string	connector_type	Indicates type of video input connector, is one of the following: HDMI for an HDMI connector. DISPLAY_PORT for a DisplayPort connector.

2.2.100 class NODE_VIDEO_COMPRESSOR : NODE

Type	Name	Description
NODE_TYPE	type	VIDEO_COMPRESSOR
VIDEO_COMPRESSOR_NODE_CONFIG	configuration	Video Compressor node configuration detail
VIDEO_COMPRESSOR_NODE_STATUS	status	Video Compressor node configuration status

2.2.101 class VIDEO_COMPRESSOR_NODE_CONFIG

Type	Name	Description
bool	always_compress	When true, HDMI stream 0 is always compressed without regards to the video format. Otherwise, compressed only when it is necessary to keep the video

		bandwidth under 9Gbit/s. Note: HDMI stream 1 is never compressed.
int	version	Video compression version to apply. Two options available: 2 For improved video compression. 1 For backward compatibility with devices that only support this version. Note: All devices receiving compressed video, HDMI stream 0 with compression enabled, must be configured to decompress with same compression version.

2.2.102 class VIDEO_COMPRESSOR_NODE_STATUS

Type	Name	Description
int	version_support	Maximum compression version supported by this device. Can be either 1 or 2.

2.2.103 class NODE_VIDEO_DECOMPRESSOR : NODE

Type	Name	Description
NODE_TYPE	type	VIDEO_DECOMPRESSOR
VIDEO_DECOMPRESSOR_NODE_CONFIG	configuration	Video Decompressor node configuration detail
VIDEO_DECOMPRESSOR_NODE_STATUS	status	Video Decompressor node configuration status

2.2.104 class VIDEO_DECOMPRESSOR_NODE_CONFIG

Type	Name	Description
int	version	Video compression version to use for decompression. Can be either: 2 Improved video compression. 1 Backwards compatibility with devices that only support this version. Must be set to same version as device sending the compressed video, such as HDMI stream 0 if compression enabled.

2.2.105 class VIDEO_DECOMPRESSOR_NODE_STATUS

Type	Name	Description
int	version_support	Maximum compression version supported by device. Can be either 1 or 2.

2.2.106class NODE_BITMAP_OVERLAY : NODE

Type	Name	Description
NODE_TYPE	type	BITMAP_OVERLAY
BITMAP_OVERLAY_NODE_CONFIG	configuration	Bitmap overlay node configuration detail

2.2.107class BITMAP_OVERLAY_NODE_CONFIG

Type	Name	Description
bool	enable	Whether bitmap overlay is enabled or disabled.
int	width	Width of the overlay bitmap.
int	height	Height of the overlay bitmap.
int	horiz_position	Horizontal position, in pixels, of the overlay
int	vert_position	Vertical position, in pixels, of the overlay
int	scaling_factor	Scaling factor (1, 2 or 4)
string	foreground_color	Foreground colour in format RRGGBB.
string	background_color	Background colour in format RRGGBB.
int	foreground_transparency	Foreground transparency level in percent.
int	background_transparency	Background transparency level in percent.

2.2.108class NODE_THUMBNAIL_GENERATOR : NODE

Type	Name	Description
NODE_TYPE	type	THUMBNAIL_GENERATOR
VIDEO_DECOMPRESSOR_NODE_CONFIG	configuration	Video Decompressor node configuration detail
VIDEO_DECOMPRESSOR_NODE_STATUS	status	Video Decompressor node configuration status

2.2.109class THUMBNAIL_GENERATOR_NODE_CONFIG

Type	Name	Description
double	frame_rate	The requested frame rate for the thumbnail stream.
bool	hdcv	Whether thumbnails are produced when video input is HDCP protected.
int	ssrc	RTP synchronization source (SSRC) identifier. This 32-bit value must be unique for each thumbnail source. See RFC 3550 section 8 for detail.
int	udp_port	Destination UDP port.

2.2.110class THUMBNAIL_GENERATOR_NODE_STATUS

Type	Name	Description
int	width	The width of each thumbnail.
int	height	The height of each thumbnail.
double	frame_rate	The actual frame rate for the thumbnail stream.
int	bits_per_comp	The value 8 which indicates that there are 8 bits per colour component.
string	color_space	The string YCBCR_422 which indicates the thumbnail stream is in the YCbCr colour space with 4:2:2 chroma subsampling.
string	colorimetry	The colorimetry of the thumbnail stream, is either BT709 for BT.709 or BT601 for BT.601

2.2.111class NODE_MULTI_LINK_TRANSMITTER : NODE

Type	Name	Description
NODE_TYPE	type	MULTI_LINK_TRANSMITTER
MULTI_LINK_TRANSMITTER_NODE_CONFIG	configuration	
MULTI_LINK_TRANSMITTER_NODE_STATUS	status	

2.2.112class MULTI_LINK_TRANSMITTER_NODE_CONFIG

Type	Name	Description
string	link_mode	

2.2.113class MULTI_LINK_TRANSMITTER_NODE_STATUS

Type	Name	Description
List<LinkCapability>	link_capabilities	List all link status capability
List<Companion>	companions	List all exist companion
string	video_framing	Specify the video frame

2.2.114class LinkCapability

Type	Name	Description
string	mode	

2.2.115class Companion

Type	Name	Description
string	device_id	The ID of Device
int	chip_id	The ID of Chipset

2.2.116class NODE_MULTI_LINK_RECEIVER : NODE

Type	Name	Description
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NODE_TYPE	type	MULTI_LINK_RECEIVER
MULTI_LINK_RECEIVER_NODE_CONFIG	configuration	Multiple link receiver node configuration detail
MULTI_LINK_RECEIVER_NODE_STATUS	status	Multiple link receiver node configuration status

2.2.117 class MULTI_LINK_RECEIVER_NODE_CONFIG

Type	Name	Description

2.2.118 class MULTI_LINK_RECEIVER_NODE_STATUS

Type	Name	Description
List<LinkCapability>	link_capabilities	List all link status capability
string	link_mode	Specify the link mode current status
List<Companion>	companions	List all exist companion
string	video_framing	Specify the video frame

2.2.119 class VIDEO

Type	Name	Description
string?	aspect_ratio	aspect argument specifies the picture aspect ratio. Two valid formats for this argument, either: 1. Directly specify the aspect ratio as a ratio, such as 4:3 or 16:9. 2. Specify one of the following: - ASPECT_NO_DATA If aspect ratio is to be left unspecified. - ASPECT_4_3 To specify a 4:3 aspect ratio. - ASPECT_16_9 To specify a 16:9 aspect ratio. - ASPECT_256_135 To specify a 256:135 aspect ratio. - ASPECT_64_27 To specify a 64:27 aspect ratio.
object?	fps	fps argument specifies the frame rate to apply in frames per second (fps). In a text command, it can be either an integer (e.g. 50) or a real number (e.g. 50.000) and can be optionally followed by the letter "m" (e.g. 60m or 30.00m) to indicate that the frame rate must be divided by 1.001 (i.e. multiplied by 1000/1001). In a REST API request, it can be either a (real or integer) number that represents the number of frames per second, or a string with the same syntax as in text commands. Note: When a receiver operates in genlock scaling or genlock wall

		mode, the frame rate must match frame rate of the video source. Device must also support specified video mode.
int?	hsync_front_porch	<code>front_porch</code> argument sets <code>hsync_front_porch</code> and <code>vsync_front_porch</code> respectively, specifying horizontal and vertical front porch, in pixels. Note: If either of these front porch arguments are specified, both must be specified. If front porch specified, <code>pulse_width</code> must also be specified.
int?	vsync_front_porch	
int?	hdmi_vic	<code>hdmi_vic</code> argument specifies HDMI Video Identification Code (HDMI_VIC). The supported values are: 1 -- 3840x2160 (2160p) 29.97Hz or 30Hz 2 -- 3840x2160 (2160p) 25Hz 3 -- 3840x2160 (2160p) 24Hz or 23.98Hz 4 -- 4096x2160 24Hz or 23.98Hz (256:135) Other values are inserted in the HDMI Vendor-Specific InfoFrame as requested but cannot be used to uniquely identify a video format.
int?	keep_width	<code>keep</code> argument sets <code>keep_width</code> and <code>keep_height</code> respectively, specifying width and height of portion of source video to be displayed (i.e. “kept”), in pixels. The <code>keep_width</code> argument must be even, that is a multiple of two. If either of these arguments is specified, both must be specified. Specific to <code>genlock wall</code> and <code>fast switch wall</code> modes and typically specified along with <code>read_offset</code> argument. Not specified for other display modes.
int?	keep_height	
int?	horiz_offset	<code>offset</code> argument sets <code>horiz_offset</code> and <code>vert_offset</code> respectively, specifying horizontal and vertical offset of displayed video from the full video source, in pixels. Specific to <code>genlock wall</code> and <code>fast switch wall</code> modes and typically is specified along with the <code>keep region size</code> argument. Not specified for other display modes.
int?	vert_offset	
string?	horiz_polarity	<code>polarity</code> argument sets <code>horiz_polarity</code> and <code>vert_polarity</code> respectively, specifying polarity of horizontal and vertical synchronization pulse. May be set to: POSITIVE -- positive sync polarity
string?	vert_polarity	

		NEGATIVE -- negative sync polarity If either of these arguments are specified, both must be specified.
int?	hsync_pulse	pulse argument sets <code>hsync_pulse</code> and <code>vsync_pulse</code> respectively, specifying width of horizontal and vertical synchronization pulse. If either of these pulse width arguments are specified, both must be specified. Note: If pulse width is specified, front porch must also be specified.
int?	vsync_pulse	
int?	width	size argument specifies <code>width</code> and <code>height</code> respectively, specifying width and height of the displayed video, in pixels. The <code>width</code> argument must be even, that is a multiple of two. If either of these pulse width arguments are specified, both must be specified. Note: Optional for multiview display mode. If not specified, video resolution set to match the multiview layout size.
int?	height	
int?	total_width	<code>total</code> argument specifies <code>total_width</code> and <code>total_height</code> arguments respectively, specifying total width and height in pixels including the blanking. If either of these arguments are specified, both must be specified.
int?	total_height	
int?	vic	<code>vic</code> argument specifies the Video Identification Code (VIC).
int?	viewport_horiz	<code>viewport</code> is a rectangular region on-screen where video is rendered, areas outside this region are black. <code>viewport viewport_horiz viewport_vert viewport_width viewport_height viewport_horiz</code> and <code>viewport_vert</code> arguments respectively specify the horizontal and vertical position of viewport's top-left corner on the screen, in pixels. <code>viewport_width</code> and <code>viewport_height</code> arguments respectively specify the width and height of the viewport, in pixels. The <code>viewport_horiz</code> and <code>viewport_width</code> arguments must be even, that is a multiple of two. The viewport must be completely contained within the screen size (Total Size). Refer to above under <code>Total Size</code> for more information.. If any of the four viewport arguments are specified, than all four must be specified. If it is left unspecified, the viewport will cover the whole screen. Note: The viewport argument is specific to the genlock wall and fast
int?	viewport_vert	
int?	viewport_width	
int?	viewport_height	

		switch wall video modes and is not applicable to any other video modes.
string?	quantization_range	quantization specifies color quantization range of the output. The following options are supported: STANDARD Appropriate quantization range for the video format as defined by the CEA-861-F standard. AUTO Same as STANDARD but explicitly state FULL or LIMITED as appropriate instead of DEFAULT in the AVI InfoFrame. This can be used to work around some monitor issues. LIMITED Force limited quantization range. FULL Force full quantization range. If this argument is omitted, the behavior is the same as if STANDARD was specified.

2.2.120class EVENT_OBJECT

Type	Name	Description
string	device_id	The ID of the device for which an event occurred. Note: This member is null for REQUEST_COMPLETE events since background command requests can target multiple devices.
int	event_id	The ID of the event.
string	event_type	The type of event, which is one of the following: DEVICE_CONNECTED Device joined the network. DEVICE_DISCONNECTED Device left the network. DEVICE_INFO_AVAILABLE Initial device data available for a newly connected device. FIRMWARE_UPDATE_COMPLETE Firmware update of device has completed. FIRMWARE_UPDATE_PROGRESS Firmware update of device in progress. FIRMWARE_UPDATE_START Firmware update started. No operations can be performed on device until update done. I2C An I2C bus master, such as a microcontroller) has generated an event through the I2C interface. INFRARED_RECEIVED Infrared data has been received. REQUEST_COMPLETE Background command complete. RS232_RECEIVED RS-232 data has been received. SETTINGS_CHANGED The settings of a device changed.

		TIP: As new versions of the API can introduce new event types, control clients should accept and ignore other event types to ensure compatibility with future versions.
int	timestamp	Time that the event occurred. The precision of this time stamp is in seconds and its origin (time 0) is unspecified.
Int?	request_id	Request ID associated with the event, null if not applicable. DEVICE_CONNECTED and DEVICE_DISCONNECTED: Request ID used to retrieve basic device data. DEVICE_INFO_AVAILABLE and SETTINGS_CHANGED: Request ID used to retrieve available data. I2C: Request ID can be used identify the source of an event. INFRARED_RECEIVED and RS232_RECEIVED: Request ID used to respectively retrieve either Infrared (IR) or RS-232 received data. REQUEST_COMPLETE: Request ID of the request that just completed.

2.2.121 class EVENT_RESULT

Type	Name	Description
List<EVENT_OBJECT>	events	List all Event object

2.2.122 class BR_EventResult

Type	Name	Description
string	status	Possible results for status are: SUCCESS: Command succeeded. ERROR: Command failed. PROCESSING: Command executed as a background command. Possible to execute other commands while waiting for result.
int ?	request_id	A unique request ID is assigned to each command executed as a background command. This member is null otherwise.
EVENT_RESULT	result	Result return value of a command. The result is set when the status reported is SUCCESS, otherwise null is reported.
ERROR	error	Error object describing reason why a command was unsuccessful. Set status is ERROR, otherwise null is reported.

2.2.123 class FIRMWARE_ITEM_OBJECT

Type	Name	Description
string	file_name	File name containing the firmware.
bool	supports_production	Indicates if firmware file format supports first-time programming (program command).

2.2.124class UPDATE_PROGRESS_OBJECT

Type	Name	Description
string	device_id	Device ID of the device from which the data was received.
int	progress	The number of progress units accomplished so far. These units are arbitrary and subject to change: they are only meaningful relative to the value of the total member.
int	total	The number of progress units for the whole operation.

2.2.125class WALL_PARAMETER

Type	Name	Description
string	WallDescription	video wall's description, no use.
int	Index	parameter Index, no use.
string	SourceMAC	TX MAC address
int	Width	source width
int	Height	source height
int	FPS	video output FPS
String[]	DestinationMAC	RX MAC address or "none"
Bool[]	IsFastSwitch	true: WALL_FAST_SWITCHED false: WALL_GENLOCKED
TIMING	Resolution	video output resolution
VIDEOWALL_ASPECT_RATIO	AspectRatio	picture aspect ratio
int	Row	video wall's horizontal display count
int	Column	video wall's vertical display count
BEZEL_PARAMETER	BezelInfo	video wall display's bezel compensation values

2.2.126class MULTIVIEW_PARAMETER

Type	Name	Description
string	MultiviewDescription	multiview's description
int	Index	parameter Index, no use.
VIDEOWALL_ASPECT_RATIO	AspectRatio	picture aspect ratio
int	LayoutID	layout ID
string	DestinationMAC	RX MAC address
TIMING	Resolution	video output resolution
String[]	SourceMAC	TX MAC address or "none"
VIDEO_INFO[]	SourceInfo	source video width and height
VIDEO_INFO[]	ScaleInfo	view size (Automatic calculation)

Byte[]	MappingID	index of surface that window referring to.
int	SourceQty	source quantity

2.2.127class BEZEL_PARAMETER

Type	Name	Description
int	Top	Video Top position parameter
int	Bottom	Video Bottom position parameter
int	Left	Video Left position parameter
int	Right	Video Right position parameter

2.2.128class VIDEO_INFO

Type	Name	Description
int	Width	Video picture Width parameter
int	Height	Video picture Height parameter

2.2.129class Multicast

Type	Name	Description
string	address	Multicast IP address in dotted decimal notation.
string	state	Current state of the stream, which is one of the following: ● STREAMING device is actively transmitting on specified multicast address. ● STOPPED the stream is stopped.
string	device_id	Device ID of device that is using the multicast address.
string	stream_type	Type of stream the multicast address is being used for.
int	stream_index	Index of stream the multicast address is being used for.

2.3 Function

2.3.1 void SET_HTTP_PORT([int Port = 80])

Parameter			Value	Description
Input	Port			Blueriver control server http port
Return				
Description	Control Server Http Port (default port 80). Define control server http port.			

2.3.2 void SET_CONTROL_SERVER(string Address, [int Port = 80])

Parameter			Value	Description
Input	string	Address		Blueriver control server IP address
	int	Port		Blueriver control server http port
Return	BR_RESULT			
Description	Define control server IP address and http port.			

2.3.3 BR_RESULT Discovery()

Parameter			Value	Description
Input				
Return	BR_RESULT			
Description	Lists devices with no information other than device ID.			

2.3.4 BR_RESULT DiscoverySubset([DEVICE_GROUPS gup = DEVICE_GROUPS.ALL])

Parameter			Value	Description
Input	DEVICE_GROUPS	gup		
Return	BR_RESULT			
Description	Retrieves basic status information regarding target device(s) that is available when first detected on the network, such as device type and firmware version.			

2.3.5 BR_RESULT DiscoveryIdentity([DEVICE_GROUPS gup = DEVICE_GROUPS.ALL])

Parameter			Value	Description
Input	DEVICE_GROUPS	gup		
Return	BR_RESULT			
Description	Retrieves basic identity information regarding target device(s), such as device name, type and firmware version.			

2.3.6 async Task<List<string>> ADD_DEVICE_ALL()

Parameter			Value	Description
Input				
Return	List<string>			Device list.
Description	Add all device.			

2.3.7 List<string> GET_DEVICE_LIST()

Parameter			Value	Description
Input				
Return	List<string>			Device list.
Description	Get device list.			

2.3.8 bool SAVE_DEVICE_LIST(string Filename)

Parameter			Value	Description
Input	string	Filename		
Return	bool			
Description	Save device list.			

2.3.9 bool LOAD_DEVICE_LIST(string Filename)

Parameter			Value	Description
Input	string	Filename		
Return	bool			
Description	Load device list			

2.3.10 BR_RESULT GET_REQUEST (int ID)

Parameter			Value	Description
Input	int	ID		The ID is a request identification number included in either: 1. A PROCESSING command response. 2. Alternatively, the associated event object.
Return	BR_RESULT			The return value is dependent on the request type.
Description	This function retrieves either: a. The return value of a background command. That is a command that responded with PROCESSING status and a request ID; b. Or, the data associated with an event. Request numbers expire 30 seconds or more after they are created.			

2.3.11 BR_EventResult QueryEvent (int AfterID, [int limit = 10])

Parameter			Value	Description
Input	int	AfterID		The event ID of the last event (event ID with the highest numerical value) retrieved by the client.
	Int	limit		The maximum number of events returned.
Return	BR_EventResult			
Description	Retrieves a list of all the events that recently occurred.			

2.3.12 BR_EventResult QueryEvent (string Type, int AfterID, [int limit = 10])

Parameter			Value	Description
Input	string	Type		
	int	AfterID		The event ID of the last event (event ID with the highest numerical value) retrieved by the client.
	Int	limit		The maximum number of events returned.
Return	BR_EventResult			
Description	Retrieves a list of all the events that recently occurred.			

2.3.13 BR_RESULT GET_ALL_SETTINGS(string MAC)

Parameter			Value	Description
Input	string	MAC		TX/RX MAC Address
Return	BR_RESULT			Device all settings
Description	Read device all settings.			

2.3.14 BR_RESULT TX_STOP_VIDEO(string MAC, int Index)

Parameter			Value	Description
Input	string	MAC		TX MAC Address
	int	Index		Index of HDMI Video stream
Return	BR_RESULT			
Description	Stop HDMI Video stream.			

2.3.15 BR_RESULT TX_START_VIDEO(string MAC, int Index)

Parameter			Value	Description
Input	string	MAC		TX MAC Address
	int	Index		Index of HDMI Video stream

Return	BR_RESULT			
Description	Start HDMI Video stream.			

2.3.16 BR_RESULT TX_DIVIDE_FRAME_RATE(string MAC, int Mode, int Value)

Parameter			Value	Description
Input	string	MAC		TX MAC Address
	int	Mode	0	on original resolution video stream
			1	on scaled resolution video stream
	int	Value	0	Disabling
			1	Enabling
Return	BR_RESULT			
Description	frame rate conversion			

2.3.17 BR_RESULT TX_DIVIDE_FRAME_RATE(string MAC, REF_TYPE Type, int Value)

Parameter			Value	Description
Input	string	MAC		TX MAC Address
	REF_TYPE	Type	HDMI	on original resolution video stream
			SCALER	on scaled resolution video stream
	int	Value	0	Disabling
			1	Enabling
Return	BR_RESULT			
Description	frame rate conversion			

2.3.18 BR_RESULT TX_AUDIO_DIRECTION(string MAC, bool IsOutput)

Parameter			Value	Description
Input	string	MAC		TX MAC Address
	bool	IsOutput	True	output
			False	input
Return	BR_RESULT			
Description	analog audio port direction			

2.3.19 BR_RESULT TX_AUDIO_SOURCE(string MAC, int Value)

Parameter			Value	Description
Input	string	MAC		TX MAC Address
	int	Value	2	local HDMI audio (stereo downmix)

			8	I2S audio subscription (audio return channel).
Return	BR_RESULT			
Description	Set audio source.			

2.3.20 BR_RESULT TX_HDCP_SUPPORT(string MAC, bool IsEnable)

Parameter			Value	Description
Input	string	MAC		TX MAC Address
	bool	IsEnable	True	Support HDCP
			False	Not support HDCP
Return	BR_RESULT			
Description	Enable/Disable HDCP Support. Some laptops and possibly other HDMI sources always encrypt video with HDCP if the sink connected to it supports it, even when not playing back HDCP-protected content. This can cause problems when trying to display video from theses HDMI sources on monitors without HDCP support. Disabling HDCP support in the transmitter device solves this issue.			

2.3.21 BR_RESULT TX_HDCP22_SUPPORT(string MAC, bool IsEnable)

Parameter			Value	Description
Input	string	MAC		TX MAC Address
	bool	IsEnable	True	Support HDCP 2.2
			False	Not support HDCP 2.2
Return	BR_RESULT			
Description	Enable/Disable HDCP 2.2 Support. When disabled, device reports to the HDMI source that only supports HDCP 1.4 (if HDCP support enabled)			

2.3.22 BR_RESULT TX_STOP_AUDIO(string MAC, int Mode)

Parameter			Value	Description
Input	string	MAC		TX MAC Address
	int	Mode	0	HDMI audio stream
			1	Single I2S (analog) audio stream
			2	Quad I2S audio stream
Return	BR_RESULT			
Description	Stop audio stream. When stopping HDMI stream, HDMI Audio stream from the same transmitter is also			

	implicitly stopped unless independent routing of HDMI audio is enabled.
--	---

2.3.23 BR_RESULT TX_STOP_AUDIO(string MAC, AUDIO_STREAM Type)

Parameter			Value	Description
Input	string	MAC		TX MAC Address
	AUDIO_STREAM	Type		
Return	BR_RESULT			
Description	Stop audio stream. When stopping HDMI stream, HDMI Audio stream from the same transmitter is also implicitly stopped unless independent routing of HDMI audio is enabled.			

2.3.24 BR_RESULT TX_START_AUDIO(string MAC, int Mode)

Parameter			Value	Description
Input	string	MAC		TX MAC Address
	int	Mode	0	HDMI audio stream
			1	Single I2S (analog) audio stream
			2	Quad I2S audio stream
Return	BR_RESULT			
Description	Start audio stream.			

2.3.25 BR_RESULT TX_START_AUDIO(string MAC, AUDIO_STREAM Type)

Parameter			Value	Description
Input	string	MAC		TX MAC Address
	AUDIO_STREAM	Type		
Return	BR_RESULT			
Description	Start audio stream.			

2.3.26 BR_RESULT TX_SET_EDID(string MAC, string Data)

Parameter			Value	Description
Input	string	MAC		TX MAC Address
	string	Data		hexadecimal string
Return	BR_RESULT			
Description	Set the EDID of transmitter device. Its length must be 512 hexadecimal characters (i.e. data length must be 256 bytes). Since the data represents an EDID, the data string must start with "00ffffffffffff00" (that is, bytes 00 ff ff ff ff ff 00, which is the EDID header).			

2.3.27 BR_RESULT RX_LEAVE_VIDEO(string MAC, int Index)

Parameter			Value	Description
Input	string	MAC		RX MAC Address
	int	Index		HDMI Index
Return	BR_RESULT			
Description	Unsubscribe receiver device from a transmitter device HDMI video stream.			

2.3.28 BR_RESULT RX_SET_GENLOCK(string MAC)

Parameter			Value	Description
Input	string	MAC		RX MAC Address
Return	BR_RESULT			
Description	Set receiver in genlock display mode.			

2.3.29 BR_RESULT RX_SET_GENLOCK_SCALING(string MAC, TIMING TimingIndex)

Parameter			Value	Description
Input	string	MAC		RX MAC Address
	TIMING	TimingIndex		
Return	BR_RESULT			
Description	Set receiver in full screen zero-frame latency scaling mode.			

2.3.30 BR_RESULT RX_SET_FAST_SWITCH_KEEP(string MAC, TIMING TimingIndex)

Parameter			Value	Description
Input	string	MAC		RX MAC Address
	TIMING	TimingIndex		
Return	BR_RESULT			
Description	Set receiver in full screen fast switch mode.			

2.3.31 BR_RESULT RX_SET_FAST_SWITCH_STRETCH(string MAC, TIMING TimingIndex)

Parameter			Value	Description
Input	string	MAC		RX MAC Address
	TIMING	TimingIndex		
Return	BR_RESULT			
Description	Set receiver in full screen fast switch stretch mode. The image is stretched to cover the entire screen without regard for the aspect			

	ratio.
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2.3.32 BR_RESULT RX_SET_FAST_SWITCH_CROP(string MAC, TIMING TimingIndex)

Parameter			Value	Description
Input	string	MAC		RX MAC Address
	TIMING	TimingIndex		
Return	BR_RESULT			
Description	Set receiver in full screen fast switch crop mode. The image is cropped to preserve the aspect ratio.			

2.3.33 BR_RESULT RX_HDMI_AUDIO_SOURCE(string MAC, int Value)

Parameter			Value	Description
Input	string	MAC		RX MAC Address
	int	Value	2	HDMI audio (original audio from video subscription in genlock mode)
			3	Analog audio
			6	HDMI audio (stereo downmix)
			7	HDMI audio (all available channels)
			8	I2S audio subscription
			9	I2S audio local loop out
Return	BR_RESULT			
Description	Set HDMI audio source.			

2.3.34 BR_RESULT MULTICH_AUDIO_SOURCE (string MAC, int Value)

Parameter			Value	Description
Input	string	MAC		MAC Address
	int	Value	3	
			6	
			7	
			8	
Return	BR_RESULT			
Description	Set audio input source.			

2.3.35 BR_RESULT RX_LEAVE_AUDIO(string MAC, int Mode)

Parameter			Value	Description
Input	string	MAC		RX MAC Address

	int	Mode	0	HDMI audio stream
			1	Single I2S (analog) audio stream
			2	Quad I2S audio stream
Return	BR_RESULT			
Description	Unsubscribe receiver device from a transmitter device audio stream.			

2.3.36 BR_RESULT RX_LEAVE_AUDIO(string MAC, AUDIO_STREAM Type)

Parameter			Value	Description
Input	string	MAC		RX MAC Address
	AUDIO_STREAM	Type		
Return	BR_RESULT			
Description	Unsubscribe receiver device from a transmitter device audio stream.			

2.3.37 BR_RESULT REBOOT(string MAC)

Parameter			Value	Description
Input	string	MAC		MAC Address
Return	BR_RESULT			
Description	Causes the target device to restart.			

2.3.38 BR_RESULT FACTORY_DEFAULT(string MAC)

Parameter			Value	Description
Input	string	MAC		MAC Address
Return	BR_RESULT			
Description	Restore the factory default settings.			

2.3.39 BR_RESULT SET_DEVICE_NAME(string MAC, string Name)

Parameter			Value	Description
Input	string	MAC		MAC Address
	string	Name	Max Length : 18	
Return	BR_RESULT			
Description	Set device's name.			

2.3.40 BR_RESULT NETWORK_SETTING(string MAC, int Mode, [string IpAddr = ""], [string Netmask = ""], [string Gateway = ""])

Parameter			Value	Description
Input	string	MAC		MAC Address
	int	Mode	0	DHCP

			1	Static
	string	IpAddr	X.X.X.X [X = 0~255]	IP address
	string	Netmask	X.X.X.X [X = 0~255]	subnet mask
	string	Gateway	X.X.X.X [X = 0~255]	network gateway
Return	BR_RESULT			
Description	Set the Internet Protocol (IP) configuration of the primary network interface of a target device.			

2.3.41 BR_RESULT SET_HELLO_STATUS(string MAC, bool OnOff)

Parameter			Value	Description
Input	string	MAC		MAC Address
	bool	OnOff	True	On
			False	Off
Return	BR_RESULT			
Description	Set hello mode on/off.			

2.3.42 BR_RESULT UART_SEND(string MAC, int Index, string Data)

Parameter			Value	Description
Input	string	MAC		MAC Address
	Int	Index		Port
	string	Data		ASCII text string.
Return	BR_RESULT			
Description	Send RS-232 data to device. This string must be escaped if it contains ASCII control characters or non-ASCII byte values. Space characters must also be escaped. The following escape sequences are recognized: \0 null character \n line feed \r carriage return \t horizontal tab \xnn (where nn are two hexadecimal characters) hexadecimal representation of byte \\ single backslash \\(space) space character			

2.3.43 BR_RESULT ROUTE_VIDEO(string SourceMAC, string DestinationMAC)

Parameter	Value	Description
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Input	string	SourceMAC		TX MAC Address
	string	DestinationMAC		RX MAC Address
Return	BR_RESULT			
Description	Subscribe receiver device to a transmitter device HDMI video stream.			

2.3.44 BR_RESULT ROUTE_VIDEO(string SourceMAC, string DestinationMAC, int StreamIndex, int SubscriptionIndex)

Parameter			Value	Description
Input	string	SourceMAC		TX MAC Address
	string	DestinationMAC		RX MAC Address
	int	StreamIndex		Index of the source HDMI stream
	int	SubscriptionIndex		Index of the destination subscription
Return	BR_RESULT			
Description	Subscribe receiver device to a transmitter device HDMI video stream.			

2.3.45 BR_RESULT ROUTE_VIDEO_GENLOCK(string SourceMAC, string DestinationMAC)

Parameter			Value	Description
Input	string	SourceMAC		TX MAC Address
	string	DestinationMAC		RX MAC Address
Return	BR_RESULT			
Description	Subscribe receiver device to a transmitter device HDMI video stream.			

2.3.46 BR_RESULT ROUTE_VIDEO_FAST_SWITCH_KEEP(string SourceMAC, string DestinationMAC, TIMING TimingIndex)

Parameter			Value	Description
Input	string	SourceMAC		TX MAC Address
	string	DestinationMAC		RX MAC Address
	TIMING	TimingIndex		
Return	BR_RESULT			
Description	Subscribe receiver device to a transmitter device HDMI video stream.			

2.3.47 BR_RESULT ROUTE_VIDEO_FAST_SWITCH_STRETCH(string SourceMAC, string DestinationMAC, TIMING TimingIndex)

Parameter			Value	Description
Input	string	SourceMAC		TX MAC Address
	string	DestinationMAC		RX MAC Address

	TIMING	TimingIndex		
Return	BR_RESULT			
Description	Subscribe receiver device to a transmitter device HDMI video stream.			

2.3.48 BR_RESULT ROUTE_VIDEO_FAST_SWITCH_CROP(string SourceMAC, string DestinationMAC, TIMING TimingIndex)

Parameter			Value	Description
Input	string	SourceMAC		TX MAC Address
	string	DestinationMAC		RX MAC Address
	TIMING	TimingIndex		
Return	BR_RESULT			
Description	Subscribe receiver device to a transmitter device HDMI video stream.			

2.3.49 BR_RESULT ROUTE_AUDIO_HDMI(string SourceMAC, string DestinationMAC)

Parameter			Value	Description
Input	string	SourceMAC		TX MAC Address
	string	DestinationMAC		RX MAC Address
Return	BR_RESULT			
Description	Subscribe receiver device to a transmitter device HDMI audio stream.			

2.3.50 BR_RESULT ROUTE_MULTICH_AUDIO (string SourceMAC, string DestinationMAC)

Parameter			Value	Description
Input	string	SourceMAC		TX MAC Address
	string	DestinationMAC		RX MAC Address
Return	BR_RESULT			
Description	Subscribe receiver device to a transmitter device I2S audio stream.			

2.3.51 void SET_VIDEO_WALL(WALL_PARAMETER Wall)

Parameter			Value	Description
Input	WALL_PARAMETER	Wall		Video wall all settings
Description	display wall mode.			

2.3.52 void SET_MULTI_VIEW(MULTIVIEW_PARAMETER Multiview)

Parameter			Value	Description
Input	MULTIVIEW_PARAMETER	Multiview		Multiview all settings
Description	Puts the device in multiview mode. Use native stream when source info is equal to window size.			

2.3.53 void SET_MULTIVIEW(MULTIVIEW_PARAMETER Multiview, [bool bTransmissionOptimized = false])

Parameter			Value	Description
Input	MULTIVIEW_PARAMETER	Multiview		Multiview all settings
	bool	bTransmissionOptimized		
Description	Puts the device in multiview mode. always use scaler stream. When the value of bTransmissionOptimized is set to true, the Display Resolution should be configured to either 4096x2160 or 3840x2160. In this scenario, the width and height of the Layout Resolution will be reduced to 75% of the Display Resolution.			

2.3.54 BR_RESULT UPDATE(string MAC, string Firmware)

Parameter			Value	Description
Input	string	MAC		MAC Address
	string	Firmware		Firmware update file
Return	BR_RESULT			
Description	updates the BlueRiver chipset firmware.			

2.3.55 BR_RESULT UPDATE(DEVICE_GROUPS gup, string Firmware)

Parameter			Value	Description
Input	DEVICE_GROUPS	gup		
	string	Firmware		Firmware update file
Return	BR_RESULT			
Description	Update the BlueRiver chipset firmware.			

2.3.56 BR_RESULT TX_AVP_SET_THUMBNAIL(string MAC, string MulticastAddress)

Parameter			Value	Description
Input	string	MAC		MAC Address

	string	MulticastAddress		Multicast Address
Return	BR_RESULT			
Description				

2.3.57 BR_RESULT TX_AVP_START_THUMBNAIL(string MAC)

Parameter			Value	Description
Input	string	MAC		MAC Address
Return	BR_RESULT			
Description	Start thumbnail stream.			

2.3.58 BR_RESULT TX_AVP_STOP_THUMBNAIL(string MAC)

Parameter			Value	Description
Input	string	MAC		MAC Address
Return	BR_RESULT			
Description	Stop thumbnail stream.			

2.3.59 BR_RESULT TX_REMOVE_HDMI_AUDIO(string TX_MAC, bool RemoveAudio)

Parameter			Value	Description
Input	string	TX_MAC		TX MAC Address
	bool	RemoveAudio		
Return	BR_RESULT			
Description	Removing Audio from HDMI Video Stream			

2.3.60 BR_RESULT TX_SET_MULTI_LINK_MODE(string TX_MAC, MULTI_LINK_MODE Mode)

Parameter			Value	Description
Input	string	TX_MAC		TX MAC Address
	MULTI_LINK_MODE	Mode		
Return	BR_RESULT			
Description	Single Mode / Dual Link Mode			

2.3.61 BR_RESULT LIST_MULTICAST()

Parameter			Value	Description
Input				
Return	BR_RESULT			
Description	Retrieves information regarding the multicast IP addresses currently in use.			

2.3.62 async Task<string> GET_SN(string MAC)

Parameter			Value	Description
Input	string	MAC		MAC Address
Return	string			Serial Number
Description	Get serial number. 1. Set the UART1 to 57600-8N1. 2. Switch UART1 to control server. 3. Send command to UART1. 4. Get EVENT RS232_RECEIVED Data. 5. Stop UART1.			

2.3.63 async Task<string> GET_MCU_IP (string MAC)

Parameter			Value	Description
Input	string	MAC		MAC Address
Return	string			MCU's IP Address
Description	Get MCU's IP Address. 6. Set the UART1 to 57600-8N1. 7. Switch UART1 to control server. 8. Send command to UART1. 9. Get EVENT RS232_RECEIVED Data. 10. Stop UART1.			

2.3.64 async Task<string> GET_MCU_VERSION (string MAC)

Parameter			Value	Description
Input	string	MAC		MAC Address
Return	string			MCU's Firmware Version
Description	Get MCU's Firmware Version. (Support PID : 30, 31, 60, 61, 80, 81, 1034, 1035, 1064, 1065, 1068, 1069) 1. Set the UART1 to 57600-8N1. 2. Switch UART1 to control server. 3. Send command to UART1. 4. Get EVENT RS232_RECEIVED Data. 5. Stop UART1.			

2.3.65 BR_RESULT TX_SET_SCALER(string MAC, int Width, int Height)

Parameter			Value	Description
Input	string	MAC		TX MAC Address

	int	Width		The width of the scaled video (in pixels)
	int	Height		The height of the scaled video (in pixels)
Return	BR_RESULT			
Description	Used to set the scaler video resolution of one transmitter device. The scaler resolution must be set before starting a scaled video stream.			

3. Class THUMBNAIL_AVP

Semtech AVX100/AVX200T and BlueRiver AVP1000/AVP2000/AVP2000T

3.1 void Start()

Parameter			Value	Description
Return				
Description	Start to listen thumbnail data.			

3.2 void Stop()

Parameter			Value	Description
Return				
Description	Stop to listen thumbnail data.			

3.3 void Close()

Parameter			Value	Description
Return				
Description	Close the UDP connect			

3.4 Bitmap GET_THUMBNAIL(string DeviceIP)

Parameter			Value	Description
Return	Bitmap			Bitmap
Description	Get a video's thumbnail.			